

REDROB AI HACKATHON 2025

SkillGraph AI

Intelligent Candidate Discovery & Ranking

A 7-layer hybrid AI pipeline that scores 100,000+ candidates with semantic matching, behavioral signals & career growth analysis.

100K+

Candidates

7

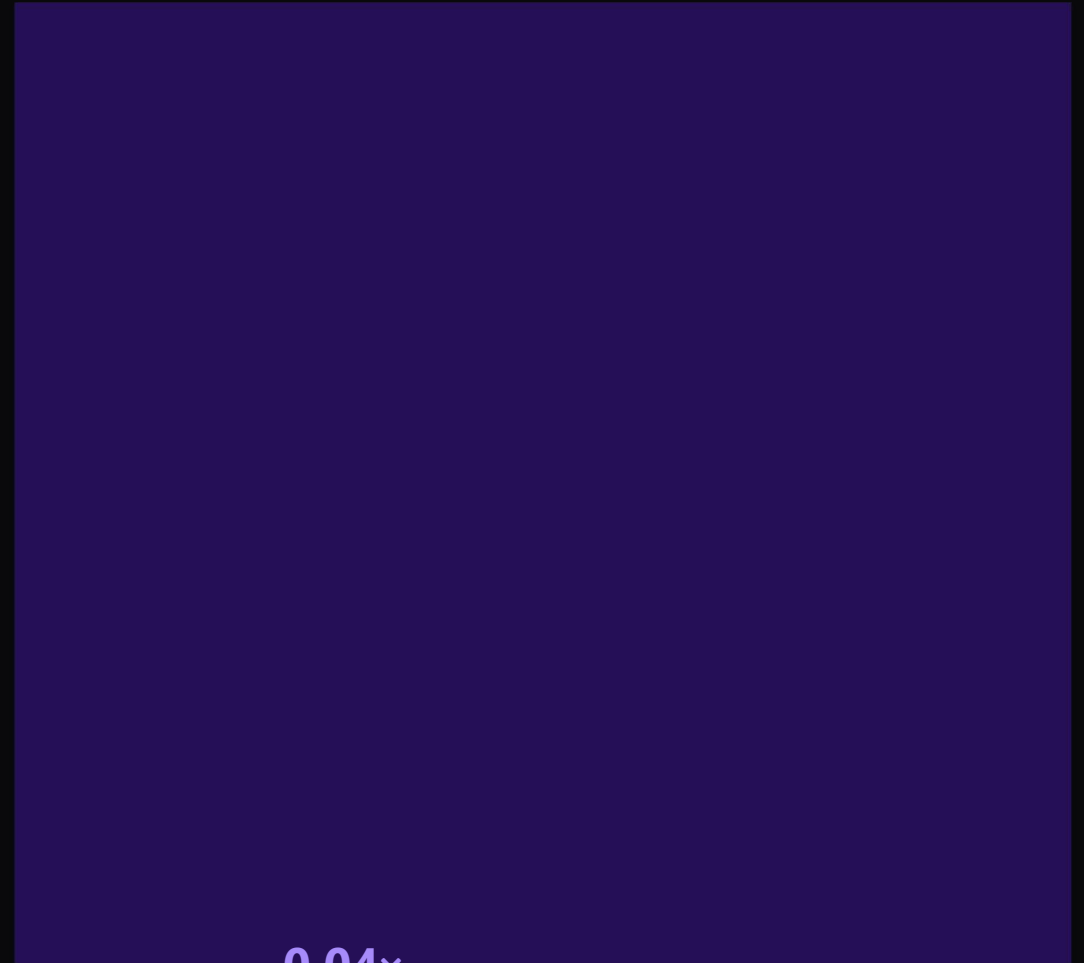
Ranking layers

<50ms

Query time

0.04x

Honeypot penalty



The Problem

Traditional hiring pipelines miss 60–80% of qualified candidates

Keyword Matching Fails

Boolean filters reject qualified candidates who use different terminology for the same skill.

Honeypot Pollution

Keyword-stuffed resumes and consulting-farm profiles inflate the apparent pool size.

Manual Review at Scale

A recruiter can review ~60 profiles/day. At 100,000 applicants, that's 1,600 working days.

Hidden Gems Lost

High-fit candidates with low online visibility are never contacted — wasted pipeline value.

Our Solution

SkillGraph: A 7-layer hybrid AI ranking pipeline

1 JD Validation	Detects impossible timelines, skill overload, contradictions
2 BM25 Lexical	Fast token-level keyword matching baseline
3 SBERT Semantic	384-dim sentence embeddings for skill equivalence
4 Experience	Years curve with AI/ML-specific weighting (peak at 7yr)
5 Project Depth	Production ML, stack complexity, open-source contributions
6 Behavior Signals	Redrob: response rate, availability, notice period
7 Career Growth	Seniority trajectory — IC vs consulting path detection

System Architecture

End-to-end inference pipeline + real-time API

Data Layer

- candidates.jsonl (100K)
- submission.csv (top-100 ranked)
- JD YAML config
- technology_timeline.yaml

AI Pipeline

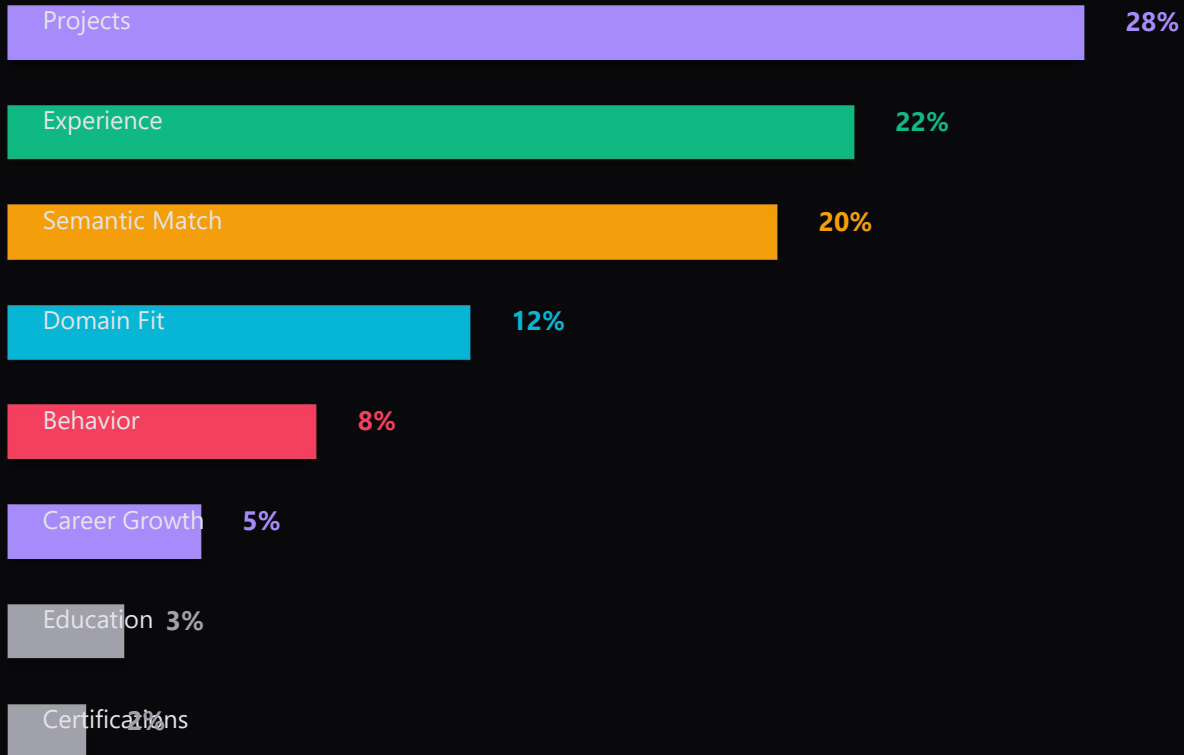
- Job Understanding Agent
- BM25 + SBERT dual-encoder
- 7-layer scorer + penalties
- Hidden Gem detector

API + Frontend

- FastAPI (15 endpoints)
- React + TypeScript + Vite
- Recharts analytics
- Real-time AI Copilot chat

Scoring Methodology

Weighted composite across 8 dimensions with multiplicative penalties



Multiplicative Penalties

- Consulting-only career ×0.40
- Honeypot profile ×0.04
- Keyword stuffer ×0.55
- Applied after weighted sum

Hidden Gem Criteria

- Score ≥ 0.63 + Career growth ≥ 0.70
- Low recruiter contact OR open_to_work
- Rank > 20 (overlooked by recruiters)

JD Intelligence Validator

Catches impossible requirements before you post them

Timeline Analysis

Detects requests like '10yr Kubernetes exp' (impossible — released 2014 → max 11yr)

Contradiction Detection

Flags conflicting signals: 'remote-only' + 'daily office standup required'

Skill Overload Scoring

Quantifies complexity → talent availability estimate → suggests priority vs optional

Missing Requirements

Identifies absent JD fields that reduce match quality (comp range, notice period, growth path)

Transferable Mapping

Maps non-standard skill names to canonical equivalents — reduces false negatives

JD Quality Score

0–1 composite score with actionable improvements applied to corrected JD profile

Pipeline Results

From 100,000 candidates to interview-ready shortlist in seconds

100K+

Total screened

12

Interview-ready

7

Hidden gems

85

Honeypots caught

100,000 Raw candidates

~8,500 Qualified (score > 0.4)

~1,200 Strong fit (score > 0.55)

~12 Interview-ready (≥ 0.70)

7 Hidden gems discovered

Top Candidate Profile (Rank #1)

- Score: 0.847 (97th percentile)
- 7.2yr total · 5.4yr AI/ML specific
- Production ML deployments: ✓
- Projects: 0.91 · Semantic: 0.83
- Notice period: 15 days
- Open to work · High response rate

Demo Flow

Natural 4-minute walkthrough of the recruiter experience

1

Landing Page

Choose Recruiter or Candidate portal

3

Candidate Rankings

Filter, search, sort 100 ranked profiles

5

Hidden Talent

7 overlooked high-fit candidates surfaced

7

AI Copilot

Natural language queries over candidate pool

2

JD Validator

Paste raw JD → see issues → quality score

4

Candidate Detail

Score breakdown, career history, signals

6

Analytics

Score distribution, funnel, skill heatmap

8

Candidate Portal

Self-service resume analysis for applicants

Technology Stack

Production-grade components, zero proprietary lock-in

AI / ML

- sentence-transformers (SBERT)
- all-MiniLM-L6-v2 (384-dim)
- BM25 (rank-bm25)
- PyYAML-driven timeline knowledge

Backend

- FastAPI + Uvicorn
- Pydantic v2 response models
- python-multipart (form uploads)
- Async + streaming-ready

Frontend

- React 18 + TypeScript + Vite
- Framer Motion (animations)
- Recharts (analytics charts)
- TanStack Query v5

Data

- JSONL candidate store (100K)
- CSV submission format
- Numpy embeddings cache
- In-memory DataStore singleton

Candidate Portal

Two-sided experience: recruiter dashboard + self-service candidate analysis

Recruiter Experience

- Real-time ranked candidate feed
- 7-dimension score breakdown per profile
- Side-by-side radar comparison (up to 5)
- JD validation with live quality score
- Hidden gem discovery panel
- Analytics: funnel, distribution, skills
- AI Copilot — natural language queries

Candidate Self-Assessment Portal

- Step 1: Browse open roles (job cards with skills)
- Step 2: Upload or paste resume text
- Step 3: Instant AI match score (0–1)
- Strengths mapped to JD categories
- Missing skills + transferable equivalents
- Personalised project recommendations
- Certification roadmap to close gaps

Key Differentiators

Why SkillGraph outperforms naive ranking approaches

Semantic Equivalence	SBERT embeddings match 'Chroma' to 'vector database' — Boolean search misses this entirely.
Honeypot Filtering	×0.04 penalty collapses scores of profiles with ≥12 rare skills + no GitHub + 0% response rate.
JD-side Intelligence	Most tools only score candidates. We also validate and improve the job description first.
Hidden Gem Detection	Explicit rule: score ≥0.63 AND growth ≥0.70 AND (low_contact OR open_to_work) AND rank>20.
Behavioral Signals	Redrob's proprietary signals (response rate, days_since_active, interview completion) integrated as scoring layer.
Career Growth Proxy	Detects IC progression vs consulting churn — penalises stagnant consulting-only trajectories.

What's Next

Path from prototype to production hiring platform

Phase 1 (Done)

- 7-layer ranking pipeline
- FastAPI backend
- React dashboard
- Candidate portal



Phase 2 (Near-term)

- FAISS ANN index for full 100K
- LLM-powered JD rewriting
- Email outreach integration
- Resume PDF parsing



Phase 3 (Scale)

- Multi-role concurrent campaigns
- Bias detection + fairness audit
- Candidate tracking CRM
- Enterprise SSO / RBAC

SkillGraph AI

Intelligent Candidate Discovery & Ranking

Built for the Redrob AI Hackathon 2025

github.com/Tanishadot/Skillgraph

Thank you